

Karim ElSedfy

Full-stack + AI/ML engineer building computer-vision, LLM-assisted, and production data systems

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EDUCATION

Bowdoin College

May 2028

B.A. in Computer Science and Mathematics

Brunswick, ME

- **GPA:** 3.85/4.00, Dean's List
- **Relevant Coursework:** Deep Learning, Artificial Intelligence, Algorithms, Computer Systems, Data Structures I & II, Linear Algebra

EXPERIENCE

Dropbox

May 2026 – Present

Software Engineer Intern

Remote - San Francisco, CA

- Ship full-stack product changes across a React/TypeScript frontend and Python backend services, using internal gRPC RPC and A/B experimentation tooling to launch and measure targeted conversion flows.
- Build onboarding and upgrade experiences on the Teams Formation team that move business-domain users on free plans toward paid Team plans, tying engineering changes to adoption and recurring revenue.

Ungated Research, Bowdoin College

Mar. 2026 – Present

Software Engineer

Brunswick, ME

- Co-maintain a multithreaded Python ETL pipeline (SQLAlchemy, MariaDB, AWS S3) ingesting economics papers from 11+ journals, enriched with Gemini-generated summaries, tags, and translations.
- Built error-detection and automated-repair diagnostics, and ran Crossref API polling across 10 years of publications to surface 6,080 new articles for ingestion.

Computer Science Department, Bowdoin College

Sep. 2025 – Dec. 2025

Independent Study Researcher

Brunswick, ME

- Researched transferring detection-trained YOLO encoders to semantic segmentation without retraining the backbone, using dense stride-1 inference, frozen-encoder decoders, and U-Net-style skip connections.
- Improved mean IoU from 0.600 to 0.684 on the CUB-200 dataset and wrote the results as an IEEE-format paper under Prof. Jeova Farias.

Palmera Systems Co.

Jun. 2025 – Aug. 2025

Software Engineer Intern

Cairo, Egypt

- Architected face-recognition and anti-spoofing for the smartIN app (8K daily users), leading dataset curation and model fine-tuning with DeepFace/OpenCV to reach 96% detection accuracy.
- Built Angular frontends and .NET Core + PostgreSQL services matching live captures in under 2s, cutting authentication latency by 50%.

PROJECTS

Persona-Motion: Audio-Driven Talking-Head Synthesis

[GitHub](#)

Python, PyTorch, Flow Matching, wav2vec2, LivePortrait, Slurm HPC

- Built an audio-driven talking-head generator that maps wav2vec2 speech features to 205-d facial-motion coefficients with a 14.7M-parameter rectified flow-matching model, rendered to video by a frozen LivePortrait animator.
- Improved lip-sync $\sim 6\times$ over an MSE regression baseline (sync-confidence 0.09 $\rightarrow$ 0.56 vs. 0.77 ground truth) using classifier-free guidance, evaluated with a from-scratch contrastive SyncNet metric and trained end-to-end on a Slurm GPU cluster.

Proteus: Virtual Try-On

[Frontend](#) | [Backend](#) | [Live Demo](#)

React, Node.js, Express, Redis, PostgreSQL, Python, PyTorch, Kubernetes, KEDA, S3

- Built an event-driven, GPU-accelerated virtual try-on system where a Redis queue and pub/sub feed a Python/PyTorch worker running CatVTON on Linode Kubernetes Engine.
- Delivered real-time product UX with WebSocket completion events, presigned S3 uploads, and backend job orchestration that lets users upload a photo and receive synthesized try-on output without page reloads.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C#

AI / Data: PyTorch, generative modeling (flow matching, CFG), OpenCV, wav2vec2, DeepFace, SQLAlchemy, Gemini-assisted pipelines

Frameworks and Libraries: React, Node.js, Express.js, Angular, .NET Core, FastAPI

Technologies and Tools: PostgreSQL, MariaDB, Redis, Docker, Kubernetes, Slurm (HPC), AWS S3, GCP, Firebase, gRPC, Git